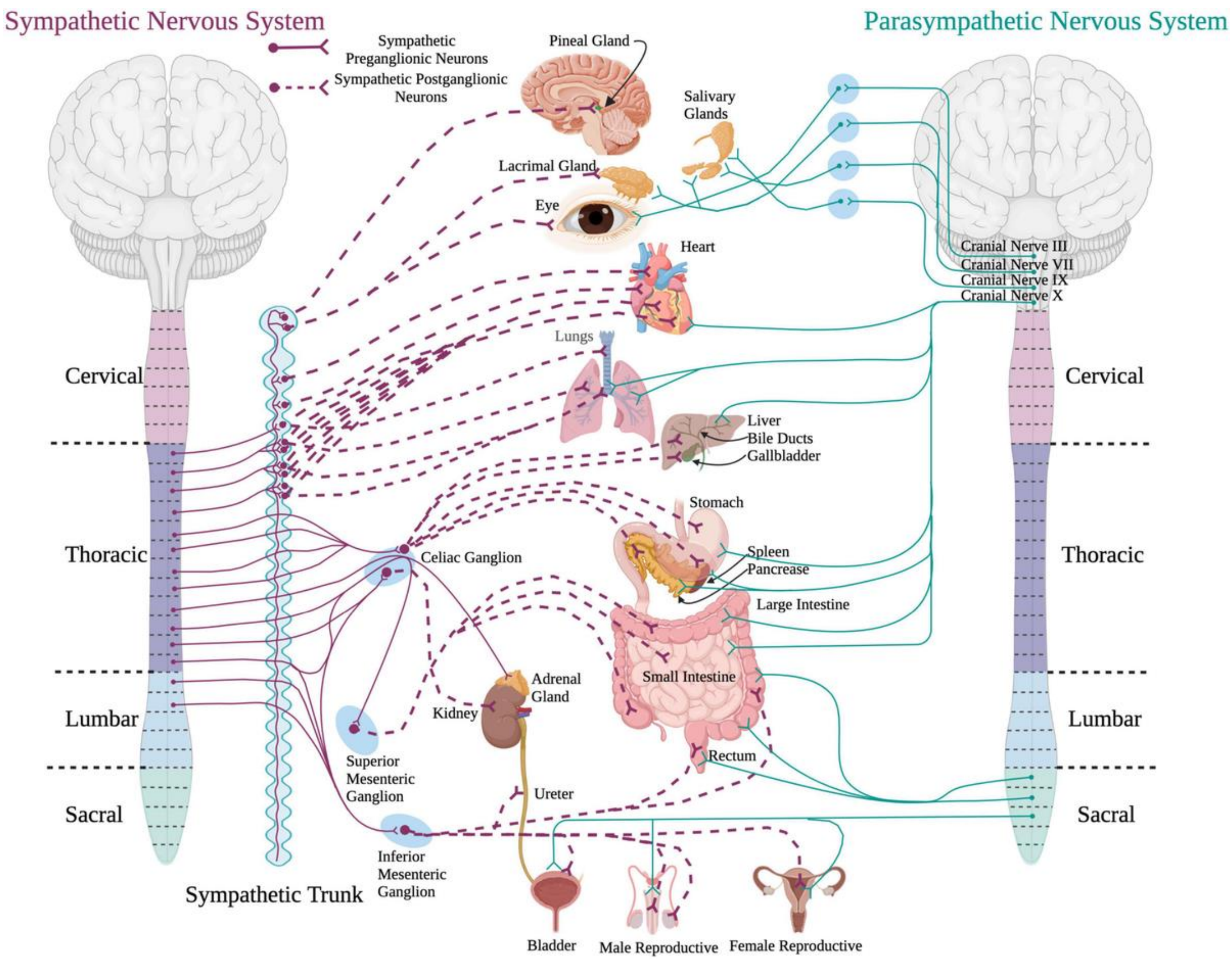
Autonomic Nervous System I

BIOL2251

Chapter 16

Sections 16.1 – 16.6



Lecture Quiz

Test what you know!

9 questions (MC + **Matching**)

Quiz automatically submits at 2:30 PM

Password = **ganglia**

Make sure you sign in before the end of lecture!

Try to use your own knowledge
as much as possible!

Building the ANS Pathways

Building the ANS Pathways

- **Instructions**
  - Work in small groups (4-6 people)
  - **Listen to each question about the anatomy of the ANS**
  - Discuss in your group for ~1-2 minutes
    - **Write your answer on your board**
  - When your group is ready, hold your board up
- **Bloom's taxonomy = Remember → Understand**

Questions

1. What is the overall function of the ANS?
Maintain homeostasis by regulating involuntary effectors (smooth muscle, cardiac muscle, glands, adipocytes)

2. What is the basic pathway of information in the ANS?
CNS (hypothalamus) → preganglionic neuron → autonomic ganglion → postganglionic neuron → effector

3. Which neurotransmitter is always released by preganglionic neurons?
Acetylcholine

Questions

4. Which neurotransmitter do sympathetic postganglionic neurons release?
Norepinephrine (except sweat glands use ACh)

5. Which neurotransmitter do parasympathetic postganglionic neurons release?
Acetylcholine

6. Write which synapses are cholinergic vs. adrenergic.
Cholinergic = all preganglionic synapses, parasympathetic postganglionic
Adrenergic = most sympathetic postganglionic neurons

Questions

7. Nicotinic receptors are found where along the ANS pathway?
All preganglionic neurons (both divisions); adrenal medulla

8. Muscarinic receptors are found in which ANS division?
Parasympathetic division (except sympathetic sweat glands)

ANS Receptor Matching

ANS Receptor Matching

- **Instructions**
  - Work in pairs
  - I will project **descriptions of different ANS receptors**
  - For each description, **identify the matching receptor name**
  - When you are ready, raise your hand to provide your answer
- **Bloom's taxonomy = Understand → Apply**

- **Receptor Options**
  - Nicotinic - Muscarinic - $\alpha 1$ - $\alpha 2$ - $\beta 1$ - $\beta 2$ - $\beta 3$

ANS Receptor Descriptions

1. Found at neuromuscular junctions and all ANS ganglia; always excitatory; ionotropic
2. Uses G-proteins; located on parasympathetic target organs; excitatory or inhibitory
3. Most common α receptor; causes excitatory responses such as vasoconstriction
4. Inhibitory α receptor; often decreases parasympathetic activity
5. Increases heart rate and contractility; generally excitatory
6. Causes bronchodilation; inhibitory effect on smooth muscle of airways
7. Found in adipose tissue; stimulates breakdown of triglycerides
8. Activated by acetylcholine (ACh)
9. Activated by norepinephrine (NE)
10. Ion channel-based receptor type that results in quick depolarization of the postsynaptic membrane

ANS Receptor Matches

Item #	Correct Match	Explanation
1	**Nicotinic**	Ionotropic, excitatory, found at NMJ and all preganglionic synapses
2	**Muscarinic**	Parasympathetic targets; G-protein; can excite/inhibit
3	**α1**	Most common α receptor, excitatory (e.g., vasoconstriction)
4	**α2**	Inhibitory; reduces parasympathetic output
5	**β1**	↑ HR, ↑ contractility; excitatory in cardiac tissue
6	**β2**	Bronchodilation; inhibitory to smooth muscle
7	**β3**	Adipose tissue; lipolysis (triglyceride breakdown)
8	**Nicotinic + Muscarinic**	Both cholinergic receptors respond to ACh
9	**α1, α2, β1, β2, β3**	All adrenergic receptors respond to norepinephrine
10	**Nicotinic**	Opens ion channels → rapid depolarization → excitation

Sympathetic vs. Parasympathetic

Sympathetic vs. Parasympathetic

- **Instructions**
  - Work in small groups (4-6 people)
  - **Complete the comparison table**
    - **Try to use only what is in your written notes and brain**
  - Turn in your table when you are finished
- **Bloom's taxonomy = Apply → Analyze → Evaluate**

Sympathetic Division

- **Origin:** Thoracolumbar (T1–L2)
- **Ganglia:** Near spinal cord/away from target (chain ganglia, collateral ganglia, adrenal medulla)
- **Preganglionic length:** Short
- **Postganglionic length:** Long
- **Neurotransmitters:** Preganglionic = ACh, Postganglionic = NE
- **Receptors:** Nicotinic, $\alpha 1$ (excitatory), $\alpha 2$ (inhibitory), $\beta 1$ (excitatory; metabolic activity + heart rate), $\beta 2$ (inhibitory; muscle relaxation in respiratory tract), $\beta 3$ (excitatory; breakdown of triglycerides in adipose tissue)
- **General Functions:** “Fight or flight,” energy mobilization, ↑ alertness

Parasympathetic Division

- **Origin:** Craniosacral (CN III, VII, IX, X + S2–S4)
- **Ganglia:** Near or within target (terminal ganglia, intramural ganglia)
- **Preganglionic length:** Long
- **Postganglionic length:** Short
- **Neurotransmitters:** Preganglionic = ACh (nicotinic), Postganglionic = ACh (muscarinic)
- **Receptors:** Nicotinic (preganglionic) and muscarinic (postganglionic)
- **General Functions:** "Rest and digest," energy conservation, nutrient storage